HOUSSAM EDDINE BEN KHEDER

Data Scientist & Applied Agentic AI Engineer

Tunis, Tunisia | +216 26 684 385 | houssameddine.benkheder@esprit.tn
linkedin.com/in/houssam-eddine-ben-kheder-101230220 | github.com/kira0987 | myportfoliohoussam-5ps9.vercel.app

PROFILE

Data Scientist and Applied Agentic AI Engineer who builds deployed RAG and agent systems that turn large document collections into reliable, decision-ready answers. Hands-on with multimodal ingestion, hybrid retrieval, and LLM agents with planning and tool use, supported by ETL, Django and FastAPI services, and MongoDB and PostgreSQL. Takes work from prototype to production.

EXPERIENCE

Data Scientist & Agentic AI Engineer — Value Digital Services
Dec 2025 – Jun 2026 | Tunis, Tunisia

- Led a production multimodal agentic RAG system for Tunisie Valeurs that answers analyst questions across 120+ equity research reports of 130+ pages each.
- Built the ingestion pipeline: YOLOv10 layout detection routes pages to GPT-4o Vision, so tables and charts become searchable.
- Built an agentic multimodal ETL that extracts and enriches image content into the database, with validation checks to ensure accuracy.
- Implemented a goal-driven agent with hierarchical planning, dynamic tool calling, multi-persona reasoning and long-term memory.
- Tuned hybrid retrieval (pgvector + PostgreSQL full-text) with multilingual-e5-large, so queries match both meaning and exact figures.
- Built an agentic CRM and advisory engine where deterministic KPI calculations back every conversational answer for Relationship Managers.
- Shipped the full stack (FastAPI, Next.js, Docker) with Laminar tracing for end-to-end agent observability.

Data Science Intern — VALUE Enterprise
Jun 2025 – Aug 2025 | Tunis, Tunisia

- Built agent-based simulations for macro and micro geomarketing analysis of high-value professional segments (lawyers, doctors, engineers).
- Built agent-based simulations using Monte Carlo methods and event-testing frameworks to model economic scenarios and market disruptions.
- Generated synthetic tabular data with CTGAN to address data scarcity; normalized client demographics concentrated in specific areas using a normal distribution; trained KNN models for customer segmentation.
- Added an agentic chatbot that explains simulation outputs to decision-makers and supports target-market conversion estimates.

Machine Learning Intern — Tritux
Jun 2024 – Aug 2024 | Tunis, Tunisia

- Built a telecom churn pipeline (2,666 customers, SMOTE+ENN balancing) benchmarking Logistic Regression, SVC, Gradient Boosting, XGBoost, AdaBoost and a neural network.
- Gradient Boosting reached 99.01% accuracy (AUC 1.00) on a 912-sample test set; SHAP explained churn drivers; K-Means found 3 customer clusters.
- Packaged the churn model as an MLOps workflow: tracked experiments, parameters and metrics in MLflow and registered the model in its model registry.
- Served predictions through a FastAPI service (/predict, /metrics) and a Flask app for browsing MLflow runs and artifacts; automated setup, training and serving with a Makefile.

KEY PROJECTS

AI-Powered Crypto Fund Due Diligence — VALUE Enterprise & ESPRIT
2024 – 2025

- Built an LLM and RAG system that scores crypto fund risks across 15 dimensions (regulatory, cybersecurity, manipulation, liquidity), stored in MongoDB.
- Processed 800+ whitepaper risk questions on GPU workstations; added LSTM volatility forecasts for BTC and ETH.
- Built a PPTX reporting agent that cut due-diligence report time from 3 days to under 10 minutes.

Deep Learning: Voice Emotion Recognition & MRI Tumor Detection — ESPRIT
2024

- Emotion recognition: trained ResNet-50 embeddings on EmoDB and RAVDESS mel-spectrograms, augmented them with diffusion-generated spectrograms, and benchmarked Logistic Regression, Random Forest and MLP classifiers (Random Forest: 98.50% test accuracy, 98.26% ± 1.91% cross-validated; MLP: 97.10%).
- Brain tumour detection: evaluated a custom CNN (92% accuracy, 0.92 macro F1) against a fine-tuned VGG16 (96% accuracy, 0.96 macro F1) on a 673-sample holdout test set; VGG16 scored higher on all four classes, most on meningioma (F1 0.83 to 0.93).

EDUCATION

Diplôme d'Ingénieur en Informatique (Master's equivalent) — ESPRIT, Tunis
Specialisation: Data Science & Software Engineering

Baccalauréat en Mathématiques — Wafa School, Ariana

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript
AI & Data: LLMs, Agentic AI, Multimodal RAG, Computer Vision, NLP, Time Series, Synthetic Data
Frameworks: LangChain, LlamaIndex, PyTorch, TensorFlow, Hugging Face, OpenCV, Scikit-learn, Next.js
Models: Transformers, GPT-4o, LLaMA, BERT, ResNet-50, YOLOv10, CTGAN, LSTM
Backend & MLOps: Django REST API, FastAPI, Docker, CI/CD, Git, Laminar
Data: PostgreSQL, MongoDB, pgvector, Vector Databases, Hybrid Search, ETL, Power BI
Languages: English (fluent), French (fluent), Arabic (native)